

Machine Learning

Assessed Coursework 2

This assignment must be submitted by June 27th, 2025 at 10:00 am.
Late submissions will be penalized by 10% **per hour of delay**.

This coursework is assessed and mandatory
and is worth 28% of your total final grade for this course.

Instructions

Identifier

Please use the random number of 6 digits you choose for the lab and include it, as a comment, in the first cell of each the notebook files you will submit – for example, “#723923”. Make sure that you keep a copy of that number as it will be used to provide the feedback (please avoid trivial numbers, such as 000000 or 123456. Also please avoid numbers starting with zero).

Submission

Submit your notebook files (compressed in zip or rar file) through EClass (name it “723923.rar” following your selected ID). The file you submit cannot be read by any other students. You can replace your submission as many times as you like by resubmitting it, although only the last version sent is kept.

If you have last-minute problems with your EClass submission, email your assignment as an attachment to alberto.paccanaro@fgv.br with the subject "URGENT – Coursework 2 SUBMISSION". In the body of the message, explain the reason for not submitting it through EClass.

All the work you submit should be solely your own work. Coursework submissions will be checked for this.

You must add your code to the given notebook file.

Note that some imports are already included in the notebook file and you are not allowed to include further ones.

All functions you need must be included in the notebook file.

In this coursework, you will implement:

1. Non-Negative Matrix Factorization (NMF) algorithm from Lee et al (1999)
2. SLIM (Sparse Linear Model) from Ning et al (2011)
3. Singular Value Decomposition (SVD)-based approach explained in class (that you can find in the textbook at page 115)

to recommend items to users. You will use the MovieLens 100K dataset, contained in the file **"U.csv"**, that you will find in EClass, that contains a matrix with ratings ranging from 1 to 5. Each of the 943 rows of the matrix corresponds to a user, and each of the 1682 columns to a movie.

EXERCISE 1 (Value: 2 marks)

You will divide your dataset into training, validation, and test sets. For each user, you will randomly take out 10 movie ratings and split them: 5 for the validation set and 5 for the test set. Your training data matrix will have the same shape as the original matrix, but the ratings of the movies that were taken out should be set to 0. Note that you need to ensure that no column in your training matrix is entirely made up of zeros.

EXERCISE 2 (Value: 7 marks)

You will implement an NMF function, that given a training matrix and the low-dimensional value k should output two non-negative matrices W and H such that their product approximates the training matrix. Your function will initialize W and H with small non-negative values and then implement a multiplicative update rule to learn the matrices. Additionally, your function should also compute the loss after each update of W and H and save it in a vector that will be returned by the function.

The training should stop when the parameters converge, that is, the relative change of the parameters is less than 0.01. Also, you should allow for a maximum of 1000 iterations.

The relative change of W can be defined as: $\frac{\max |W_{old} - W_{new}|}{\max(W_{old})}$.

EXERCISE 3 (Value: 7 marks)

You will implement a SLIM function, that given a training matrix, a learning rate, and regularization hyperparameters should output a non-negative matrix M such that the product of the training matrix with M approximates the training matrix. Your function will initialize M with small non-negative values and then implement gradient descent with projection to learn M (projection are necessary to keep the values of M positive, and this is done by simply setting the negative values in M to zero after each update). Additionally, your function should compute the loss after each update of M and save it in a vector that will be returned by the function.

The training should stop when the parameters converge, that is, the relative change of the parameters is less than 0.01. Also, you should allow for a maximum of 1000 iterations.

EXERCISE 4 (Value: 7 marks)

You will implement Singular Value Decomposition (SVD)-based approach explained in class, that given a training matrix should output the predictions of the SVD recommendation algorithm. Additionally, your function should also compute the loss after each update of the factor matrices and save it in a vector that will be returned by the function.

The training should stop when the parameters converge, that is, the relative change of the parameters is less than 0.01. Also, you should allow for a maximum of 1000 iterations.

EXERCISE 5 (Value: 5 marks)

Use your functions to evaluate the performance of NMF, SLIM and SVD at recommending movies to users. For each user, you will predict the rating of the movies that were not rated in the training set. Then you will evaluate your predictions using the root mean square error (RMSE). The RMSE should be calculated between the known ratings in the validation/test set and the predictions made by your model.

Additionally, you will compute the recall at top 10 across all users. That is, for each user, the proportion of movies with high ratings (4 and 5) that are found at the top 10 recommendation given by the prediction of a model. You will report the mean of this recall across all users. Note that users with no ratings of 4 or 5 in the test set should be excluded from this calculation.

Note that, in each exercise:

- Points will be given for choosing the model hyperparameters using a validation set.
- Include bar plots to show your results, and also a figure of the loss for each of the implemented recommender systems. Your plots should be descriptive, including titles, axis labels, and legends.